# Error Sources

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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## 1. Error is not a mistake

In careful measurement language, distinguish three things. A **mistake** (reading the wrong star, transcribing a number incorrectly, using the wrong filter) is avoidable and should be corrected. A **known systematic offset** is a bias that should be calibrated out. **Uncertainty** is the quantified doubt that remains after corrections; it is unavoidable. Older texts often say “error” for this residual uncertainty, but the words are not interchangeable.

The two broad families are random error and systematic error. Random error causes scatter; systematic error causes an offset. A measurement can be precise but inaccurate, or accurate on average but noisy.

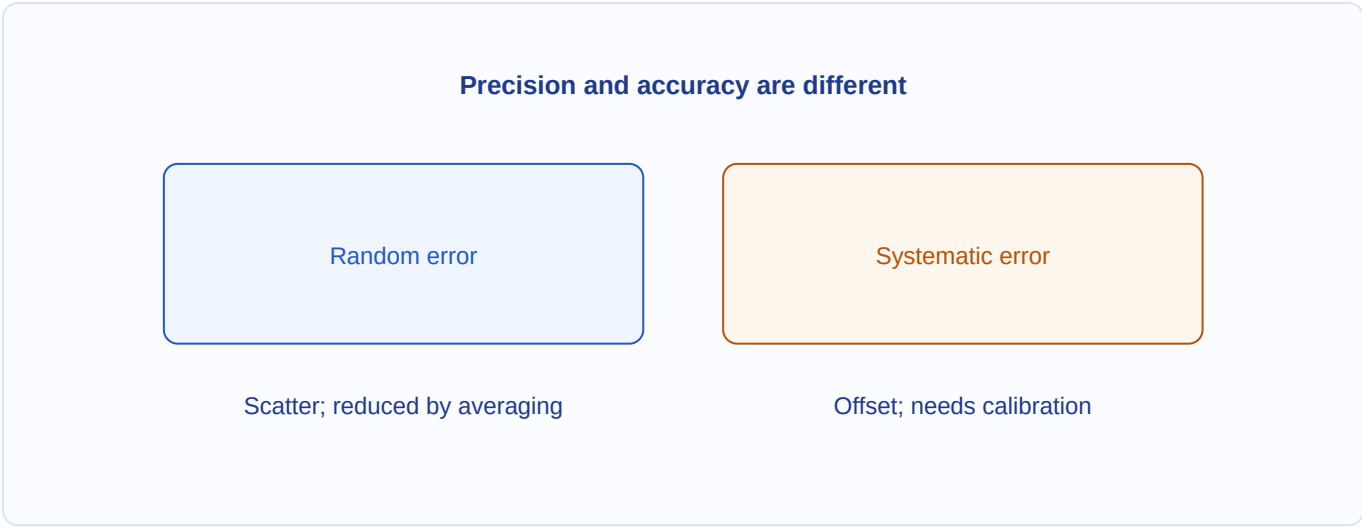



**Figure 1.** A precise measurement can still be inaccurate.

## 2. Random error and averaging

Random errors arise from photon statistics, atmospheric turbulence, electronic noise and small reading variations. If they are independent and have standard deviation $\sigma$, the standard error of the mean of $n$ readings is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}. \tag{1}$$

Averaging helps, but slowly: to halve the random uncertainty you need four times as many readings. Averaging does not help if the error is correlated, such as a slowly drifting transparency or a fixed calibration offset.

## 3. Systematic error and calibration

Systematic errors include a wrong zero point, flat-field error, atmospheric extinction, colour terms, clock error, ruler zero error and model mismatch. They do not average away. They must be calibrated, modelled or bounded.

A good calibration uses an independent standard: a known star, a known wavelength, a known length, a known time. The uncertainty in the standard becomes part of the measurement uncertainty.

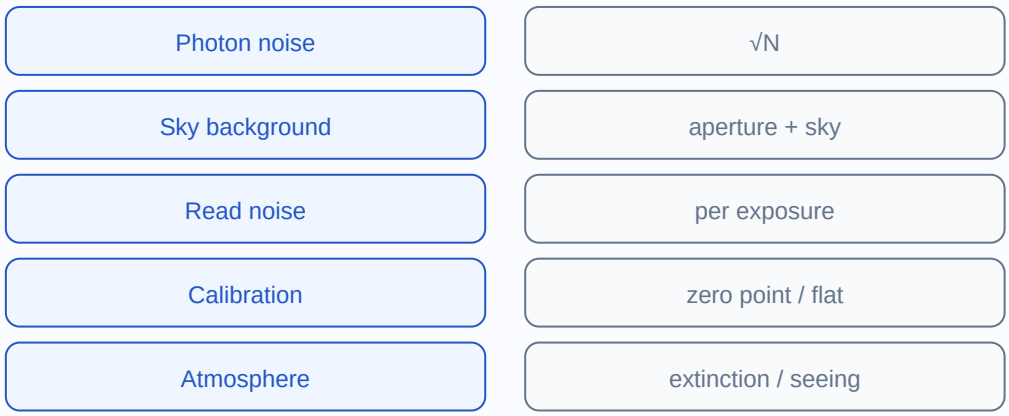



**Figure 2.** Quadrature applies to independent standard uncertainties; correlated terms need covariance.

## 4. Common astronomical error sources

Photon noise scales as $\sqrt{N}$. Sky background adds noise and depends on aperture size and sky brightness. Read noise matters for short exposures. Dark current matters for long exposures. Seeing changes the fraction of light in a photometric aperture. Extinction depends on airmass and colour. Tracking errors trail images. Timing errors shift phases. Catalogue errors affect astrometry and photometric zero points.

## 5. Building an error budget

List every term, estimate its size, correct known offsets, and combine independent, zero-mean standard uncertainties in quadrature. Identify the dominant term. If calibration dominates, more exposure time will not help much. If photon noise dominates, a larger aperture or longer exposure will help. If systematics dominate, better calibration or a different method is needed.

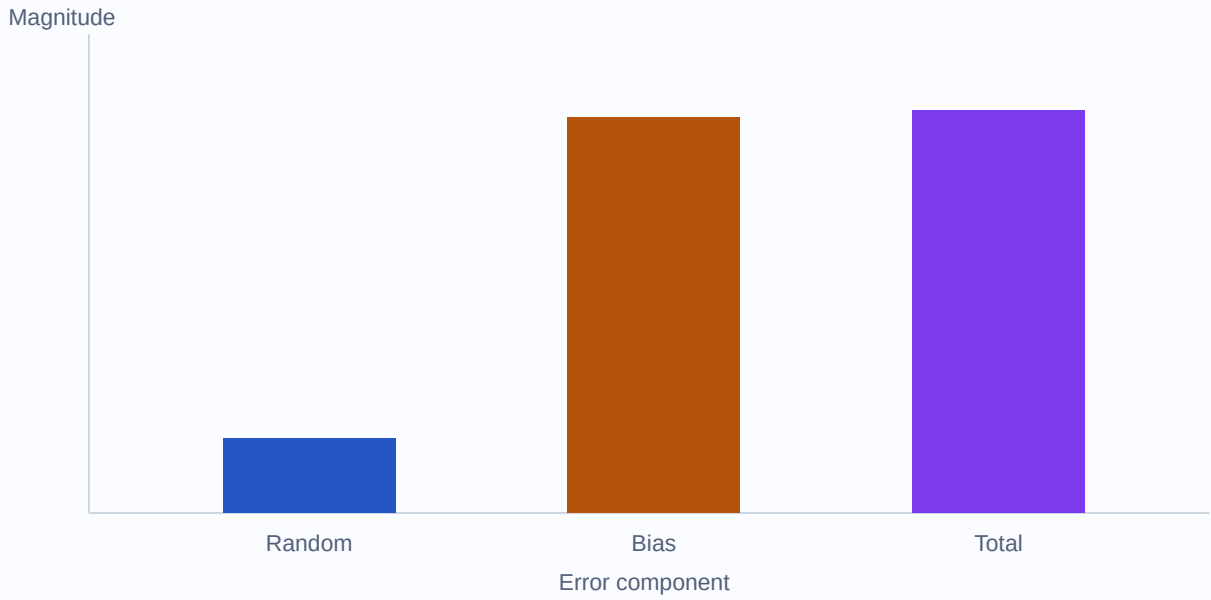
INTERACTIVE EXPERIMENT

### Averaging beats random noise, not a wrong zero.

Repeated readings contain a fixed bias plus random scatter. The mean reduces random uncertainty as 1/√n but leaves a known bias unchanged; correct the bias before combining residual uncertainties.

| Expected mean | Random uncertainty | RSS if bias is an uncertainty |
|---|---|---|
| **10.500** | **0.095** | **0.509** |

| Remaining bias |
|---|
| **0.500** |




Random uncertainty falls as 1/√n. A known systematic offset should be corrected, not averaged away. The RSS bar is illustrative only if the bias is treated as a standard uncertainty.

**Try this.** Increase n to 100. Which part of the error shrinks? Set the bias to zero and repeat.

Snapshot of the experiment. Open the HTML edition to change the controls.

## 6. Reporting uncertainty

Report the value, uncertainty, units and method. State whether the uncertainty is random, systematic or combined. If a result disagrees with a prediction, compare the difference with the combined uncertainty before claiming a discovery. A 2σ discrepancy is interesting; a 0.2σ discrepancy is not.

***A reliable method.*** Ask: What would make this measurement wrong in the same direction every time? Calibrate that. Ask: What makes it scatter? Average that. Then report both.

## 7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

**[1] Problem 1 — Random versus systematic.** Classify each as mainly random or systematic: photon noise, a ruler zero error, atmospheric scintillation, a wrong filter zero point.

Worked solution →

**[1] Problem 2 — Averaging.** A single reading has $\sigma = 0.3$. What is the standard error of the mean of 9 readings? How many readings are needed to reach 0.05?

Worked solution →

**[2] Problem 3 — Bias does not average.** A scale reads 0.2 kg too high. You average 100 measurements. What happens to the bias? What is the correct remedy?

Worked solution →

**[2] Problem 4 — Combined uncertainty.** A measurement has random uncertainty 0.03 mag and calibration uncertainty 0.04 mag. Estimate the combined uncertainty if they are independent.

Worked solution →

**[2] Problem 5 — Photon noise.** A source produces 10,000 detected photons. Estimate the photon-noise uncertainty and fractional uncertainty.

Worked solution →

**[1] Problem 6 — Precision and accuracy.** A set of measurements is tightly clustered but far from the true value. Is it precise, accurate, both or neither? What kind of error is likely?

Worked solution →

**[2] Problem 7 — Extinction error.** An extinction coefficient is uncertain by 0.02 mag/airmass. At airmass 2, what magnitude uncertainty does this introduce?

Worked solution →

**[2] Problem 8 — Timing error.** A periodic signal has period 2 hours. A timing error of 1 minute corresponds to what phase error?

Worked solution →

**[3] Problem 9 — Dominant term.** An error budget has photon noise 0.01 mag, sky noise 0.02 mag, flat-field error 0.05 mag and zero-point error 0.02 mag. Which term dominates? What should be improved first?

Worked solution →

**[3] Problem 10 — Discrepancy.** A measured value is $10.5 \pm 0.2$ and a prediction is 10.0. Is this a significant discrepancy? What if the uncertainty were 0.5?

Worked solution →

## Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

***Sources and scope.*** Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.